Editor

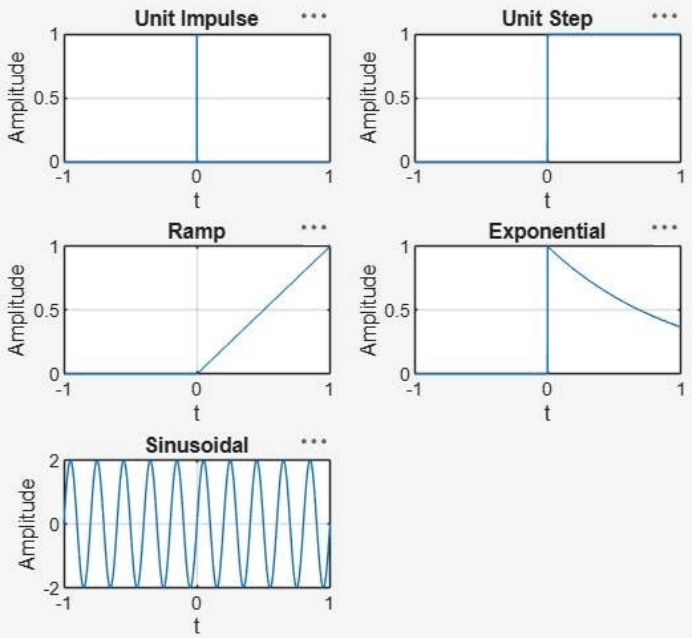
exp_1_signals.m

/MATLAB Drive/exp_1_signals.m

```
clc; clear; close all;
t = -1:0.001:1;
A = 2; f0 = 5; a = 1;
imp = double(abs(t)<0.0005);
step = double(t>=0);
ramp = t.*step;
expSig = exp(-a*t).*step;
sinSig = A*sin(2*pi*f0*t);
figure;
subplot(3,2,1); plot(t,imp); grid on; title('Uni
subplot(3,2,2); plot(t,step); grid on; title('Un
subplot(3,2,3); plot(t,ramp); grid on; title('Ra
subplot(3,2,4); plot(t,expSig); grid on; title('
subplot(3,2,5); plot(t,sinSig); grid on; title('
```

Figures

Figure 1



Command Window

/ › MATLAB Drive

Editor

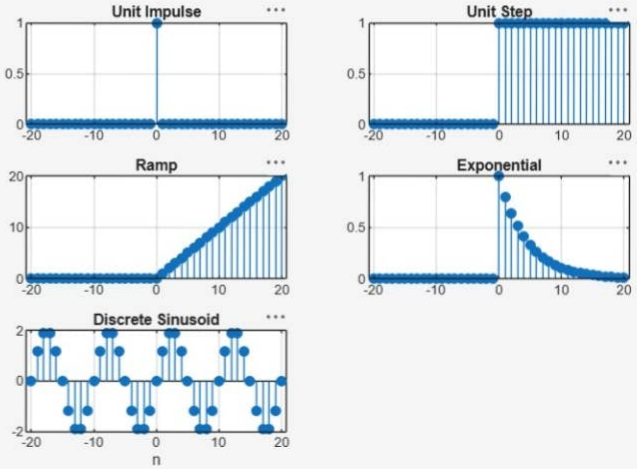
exp_1_signals.m × | exp_1_2.m × | exp2_amplitude.m × | exp2_add.m × | exp_3.m × | exp_4.m

/MATLAB Drive/exp_1_2.m

```
clc; clear; close all;
n = -20:20;
A = 2; f0 = 0.1; a = 0.1;
imp = double(n==0);
step = double(n>=0);
ramp = n.*step;
expSig = (0.8).^n .* step;
sinSig = A*sin(2*pi*f0*n);
figure;
subplot(3,2,1); stem(n,imp,'filled'); grid on; title('Unit Impulse');
subplot(3,2,2); stem(n,step,'filled'); grid on; title('Unit Step');
subplot(3,2,3); stem(n,ramp,'filled'); grid on; title('Ramp');
subplot(3,2,4); stem(n,expSig,'filled'); grid on; title('Exponential');
subplot(3,2,5); stem(n,sinSig,'filled'); grid on; title('Discrete Sinus
xlabel('n');
```

Figures

Figure 1 ×



MATLAB Drive

Editor

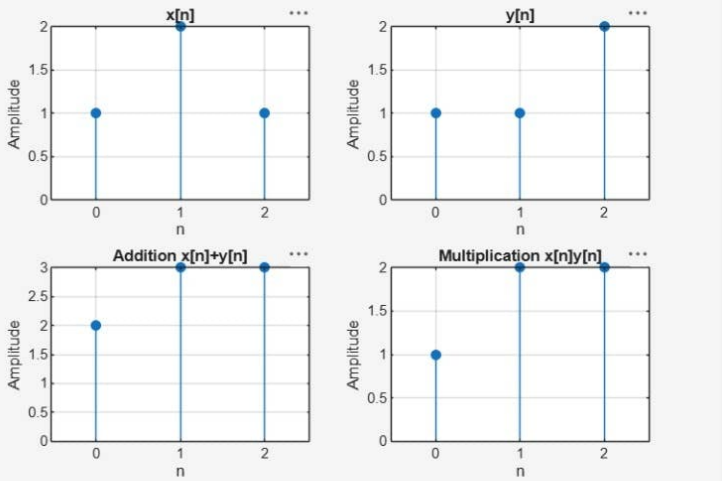
exp_1_signals.m | exp_1_2.m | exp2_amplitude.m | exp2_add.m | exp_3.m | exp_4.m

/MATLAB Drive/exp2_add.m

```
clc; clear; close all;
n = 0:2;
x = [1 2 1];
y = [1 1 2];
zSum = x + y;
zProd = x .* y;
figure;
subplot(2,2,1); stem(n,x,'filled'); grid on;
title('x[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,2); stem(n,y,'filled'); grid on;
title('y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,3); stem(n,zSum,'filled'); grid on;
title('Addition x[n]+y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,4); stem(n,zProd,'filled'); grid on;
title('Multiplication x[n]y[n]'); xlabel('n'); ylabel('Amplitude');
```

Figures

Figure 1



Command Window

New to MATLAB? See resources for Getting Started.

ATLAB Drive

Editor

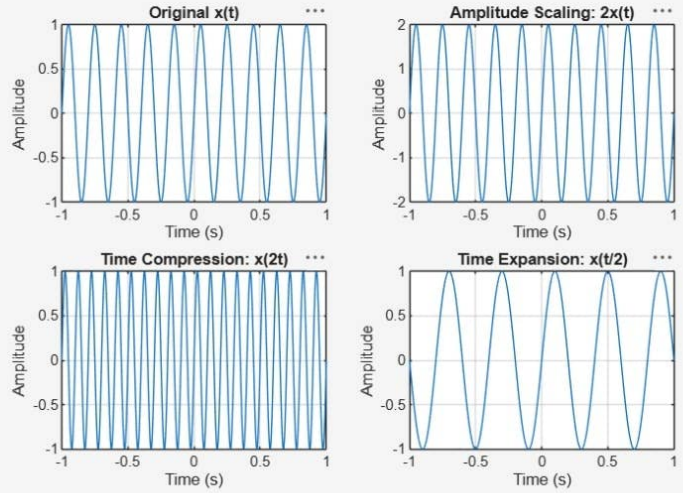
exp_1_signals.m | exp_1_2.m | exp2_amplitude.m | exp2_add.m | exp_3.m | exp_4.1

/MATLAB Drive/exp2_amplitude.m

```
clc; clear; close all;
t = -1:0.001:1;
x = sin(2*pi*5*t);
yA = 2*x;
yC = sin(2*pi*5*(2*t));
yE = sin(2*pi*5*(t/2));
figure;
subplot(2,2,1); plot(t,x); grid on;
title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,2); plot(t,yA); grid on;
title('Amplitude Scaling: 2x(t)'); xlabel('Time (s)'); ylabel('Amplitud
subplot(2,2,3); plot(t,yC); grid on;
title('Time Compression: x(2t)'); xlabel('Time (s)'); ylabel('Amplitude
subplot(2,2,4); plot(t,yE); grid on;
title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude'
```

Figures

Figure 1



Command Window

Editor

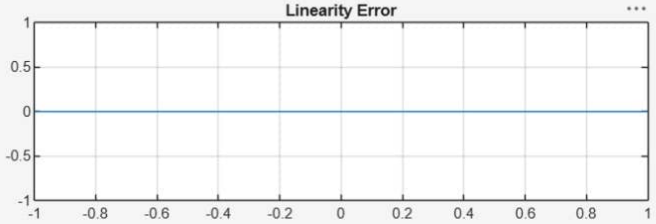
exp_1_2.m | exp2_amplitude.m | exp2_add.m | exp_3.m | exp_4.m | exp_5_1.m

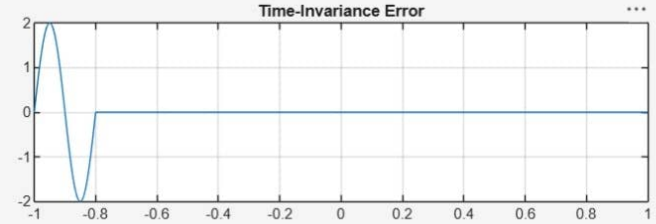
/MATLAB Drive/exp_4.m

```
clc; clear; close all;
t = -1:0.001:1;
x1 = sin(2*pi*5*t); x2 = cos(2*pi*5*t);
a = 2; b = 3; t0 = 0.2;
T = @(x) 2*x;
lhs = T(a*x1+b*x2);
rhs = a*T(x1)+b*T(x2);
linError = max(abs(lhs-rhs));
xDelay = sin(2*pi*5*(t-t0));
lhsTI = T(xDelay);
rhsTI = T(x1);
rhsTIshift = interp1(t,rhsTI,t-t0,'linear',0);
tiError = max(abs(lhsTI-rhsTIshift));
figure;
subplot(2,1,1); plot(t,lhs-rhs); grid on; title('Linearity Error');
subplot(2,1,2); plot(t,lhsTI-rhsTIshift); grid on; title('Time-Invarian
fprintf('Linearity error = %g\n',linError);
fprintf('Time-invariance error = %g\n',tiError);
```

Figures

Figure 1





Editor

exp_1_signals.m | exp_1_2.m | exp2_amplitude.m | exp2_add.m | exp_3.m | exp_4.1

/MATLAB Drive/exp_3.m

```
clc; clear; close all;
dt = 0.001;
t = 0:dt:5;
xE = exp(-t);
xP = sin(2*pi*5*t);
E_num = trapz(t,abs(xE).^2);
T0 = 1/5;
idx = t <= 2;
P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
    trapz(t(idx),abs(xP(idx)).^2);
figure;
subplot(2,1,1);
plot(t,xE); grid on;
title('Energy Signal e^{-t}u(t)');
xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,1,2);
plot(t,xP); grid on;
title('Power Signal sin(2pi5t)');
xlabel('Time (s)'); ylabel('Amplitude');
fprintf('Numerical energy = %g\n',E_num);
fprintf('Numerical average power = %g\n',P_num);

```

Figures

Figure 1

